

Section 1: Stack Implementation and Basic Operations (Easy)

These problems build your foundation — implementing a stack, using it for simple simulation, and understanding its LIFO nature.

Q1. Implement Stack using Arrays / Linked List Build a stack from scratch supporting push, pop, peek, and isEmpty operations.  GFG:

<https://www.geeksforgeeks.org/implement-a-stack-using-singly-linked-list/>

Q2. Implement Stack using Queues Implement a stack using only two queues. Try to make either push or pop $O(1)$.  LeetCode:

<https://leetcode.com/problems/implement-stack-using-queues/>

Q3. Valid Parentheses Given a string with brackets, determine if the brackets are correctly matched and nested.  LeetCode: <https://leetcode.com/problems/valid-parentheses/>

Q4. Reverse a String using Stack Use a stack to reverse a given string without using any built-in reverse function.  GFG:

<https://www.geeksforgeeks.org/stack-set-3-reversing-string-using-stack/>

Q5. Implement Two Stacks in an Array Design a data structure to implement two stacks in a single array efficiently without wasting space.  GFG:

<https://www.geeksforgeeks.org/implement-two-stacks-in-an-array/>

Section 2: Parentheses and Expression Problems (Easy-Medium)

These problems use stacks to validate, score, or evaluate expressions and bracket sequences.

Q1. Minimum Remove to Make Valid Parentheses Remove the minimum number of parentheses to make the input string valid. Return the result string.  LeetCode:

<https://leetcode.com/problems/minimum-remove-to-make-valid-parentheses/>

Q2. Score of Parentheses Given a balanced parentheses string, compute its score where $() = 1$, and $AB = A+B$, and $(A) = 2*A$.  LeetCode:

<https://leetcode.com/problems/score-of-parentheses/>

Q3. Evaluate Reverse Polish Notation Evaluate an arithmetic expression given in postfix (RPN) notation.  LeetCode: <https://leetcode.com/problems/evaluate-reverse-polish-notation/>

Q4. Basic Calculator II Implement a basic calculator to evaluate a string expression with +, -, *, / and spaces. 🔗 LeetCode: <https://leetcode.com/problems/basic-calculator-ii/>

Q5. Longest Valid Parentheses Find the length of the longest valid (well-formed) parentheses substring. 🔗 LeetCode: <https://leetcode.com/problems/longest-valid-parentheses/>

Section 3: Monotonic Stack — Next Greater / Smaller (Medium)

These problems use a monotonic stack — a stack that maintains elements in increasing or decreasing order — to find next/previous greater or smaller elements efficiently.

Q1. Next Greater Element I For each element in nums1, find the next greater element in nums2. Return -1 if none exists. 🔗 LeetCode: <https://leetcode.com/problems/next-greater-element-i/>

Q2. Next Greater Element II (Circular Array) Same as above but the array is circular — after the last element, it wraps back to the first. 🔗 LeetCode: <https://leetcode.com/problems/next-greater-element-ii/>

Q3. Previous Smaller Element For each element, find the nearest smaller element to its left. Return -1 if none exists. 🔗 GFG: <https://www.geeksforgeeks.org/previous-smaller-element/>

Q4. Stock Span Problem For each day's stock price, find the span — the number of consecutive days (including today) where the price was less than or equal to today's price. 🔗 GFG: <https://www.geeksforgeeks.org/the-stock-span-problem/> 🔗 LeetCode (Online Stock Span): <https://leetcode.com/problems/online-stock-span/>

Q5. Sum of Subarray Minimums Find the sum of the minimum element of every subarray of a given array. Use monotonic stack to achieve $O(n)$. 🔗 LeetCode: <https://leetcode.com/problems/sum-of-subarray-minimums/>

Section 4: Stack on Histograms and 2D Problems (Hard)

These problems apply the monotonic stack on histogram bars and extend the idea to 2D matrices.

Q1. Largest Rectangle in Histogram Find the area of the largest rectangle that can be formed in a histogram. 🔗 LeetCode: <https://leetcode.com/problems/largest-rectangle-in-histogram/>

Q2. Maximal Rectangle Given a binary matrix, find the largest rectangle containing only 1s. Extend the histogram approach row by row.  LeetCode: <https://leetcode.com/problems/maximal-rectangle/>

Q3. Trapping Rain Water Given an elevation map, compute how much water it can trap after raining. Solve using stack and also using two pointers.  LeetCode: <https://leetcode.com/problems/trapping-rain-water/>

Q4. Maximum Rectangle in Binary Matrix with all 1s Find the largest rectangular sub-matrix whose transpose contains all 1s.  GFG: <https://www.geeksforgeeks.org/maximum-size-rectangle-binary-sub-matrix-1s/>

Q5. Buildings With an Ocean View Find all buildings that have a clear ocean view — no taller building exists to their right.  LeetCode: <https://leetcode.com/problems/buildings-with-an-ocean-view/>

Section 5: Special Stack Designs (Hard)

These problems require augmenting or redesigning the stack to support additional operations efficiently.

Q1. Min Stack Design a stack that supports push, pop, top, and retrieving the minimum element in O(1) time.  LeetCode: <https://leetcode.com/problems/min-stack/>

Q2. Max Stack Design a stack that supports push, pop, top, peekMax, and popMax operations.  LeetCode: <https://leetcode.com/problems/max-stack/>

Q3. Design a Stack with Increment Operation Design a stack supporting push, pop, and an increment operation that increments the bottom k elements by val.  LeetCode: <https://leetcode.com/problems/design-a-stack-with-increment-operation/>

Q4. Flatten Nested List Iterator Implement an iterator for a nested list of integers where each element is either an integer or a list. Use a stack internally.  LeetCode: <https://leetcode.com/problems/flatten-nested-list-iterator/>

Q5. Decode String Given an encoded string like 3[a2[bc]], decode it to aaabcbcaabcbcb. Use two stacks — one for counts, one for strings.  LeetCode: <https://leetcode.com/problems/decode-string/>